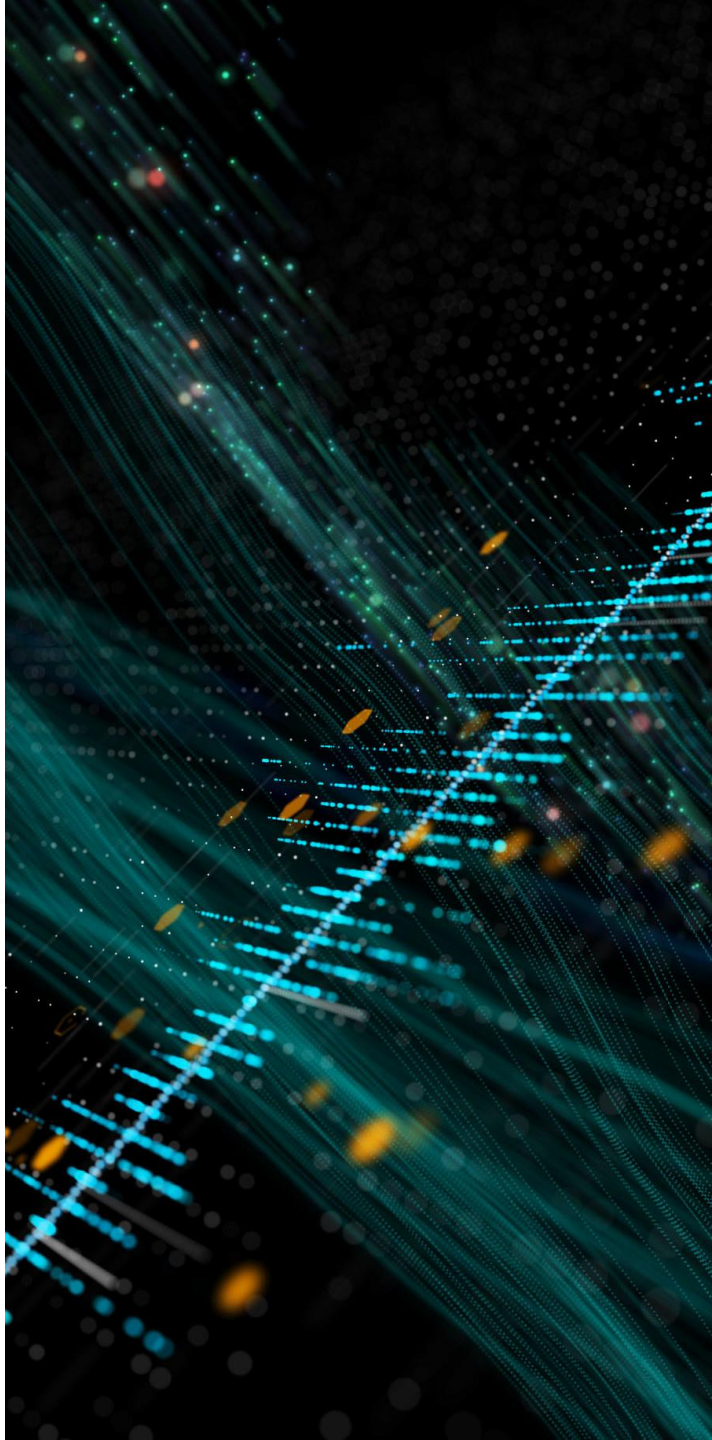
Linear Time Invariant (LTI) Systems

Semester 2, 2025/2026



Recap from Week 1

Linearity

Weighted sum of inputs $\rightarrow$ *Weighted sum of outputs*

$$ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t), \quad \forall a, b \in \mathbb{C}$$

Time-invariance

Time shift in input $\rightarrow$ *Identical time shift in output*

$$x(t) \rightarrow y(t) \Rightarrow x(t - t_0) \rightarrow y(t - t_0)$$

Exercise

Which of the following continuous-time systems are:

(i) linear

(ii) time-invariant?

1. $y(t) = \cos(3t)\, x(t)$
2. $y(t) = 3x^2(t)$
3. $y(t) = 3x(t) + 5$

Answer:
(i) Only system (1) is linear
(ii) Systems (2) & (3) are time-invariant

Exercise

Which of the following discrete-time systems are:

(i) linear

(ii) time-invariant?

1. $y[n] = x[-n]$
2. $y[n] = nx[n]$
3. $y[n] = x[n-2] - 2x[n-8]$

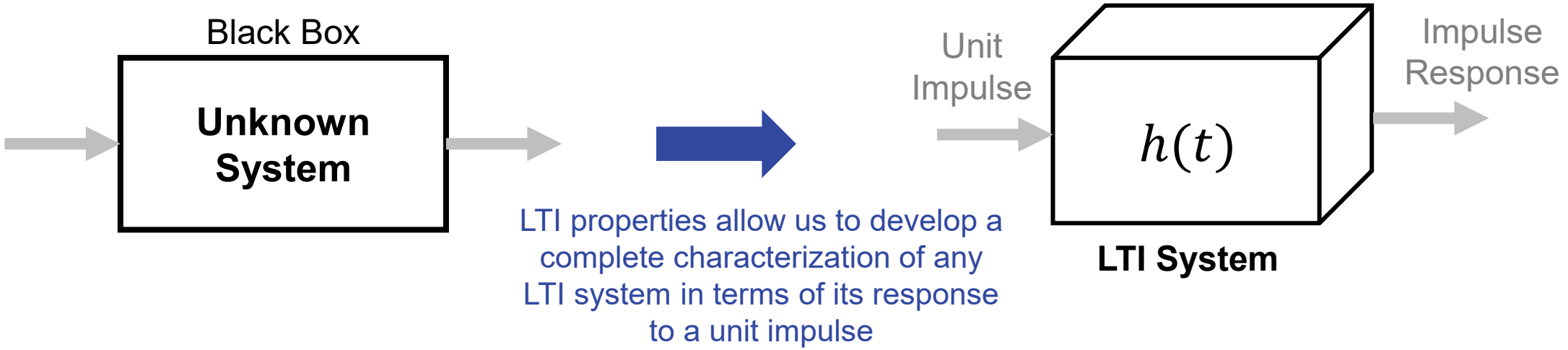
Answer:

(i) All systems are linear

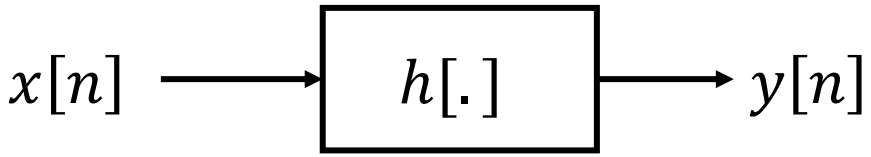
(ii) Only system (3) is time-invariant

The Power of LTI Systems

If a system is **linear** and **time-invariant**, we can predict its response to any input based on its response to a single elementary signal.



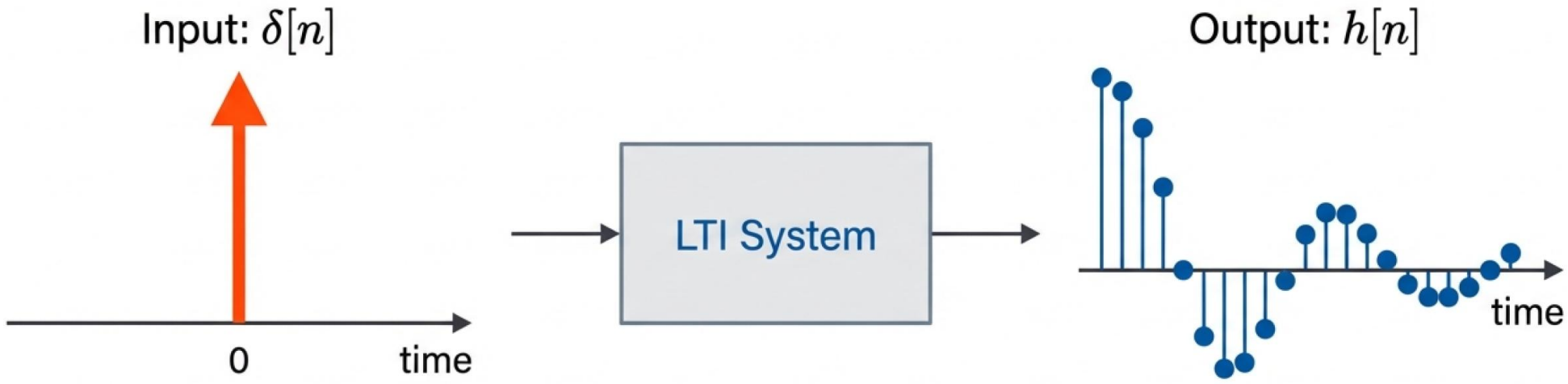
Indeed, the impulse response $h[n]$ completely characterizes the LTI system S, so you often see



Impulse Response (h)

The impulse response is the output of an LTI system when the input is a unit impulse.

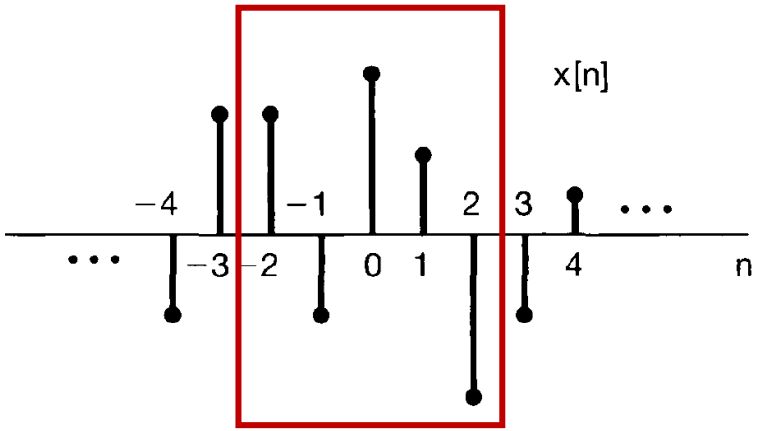
- Notation: $h[n]$ or $h(t)$



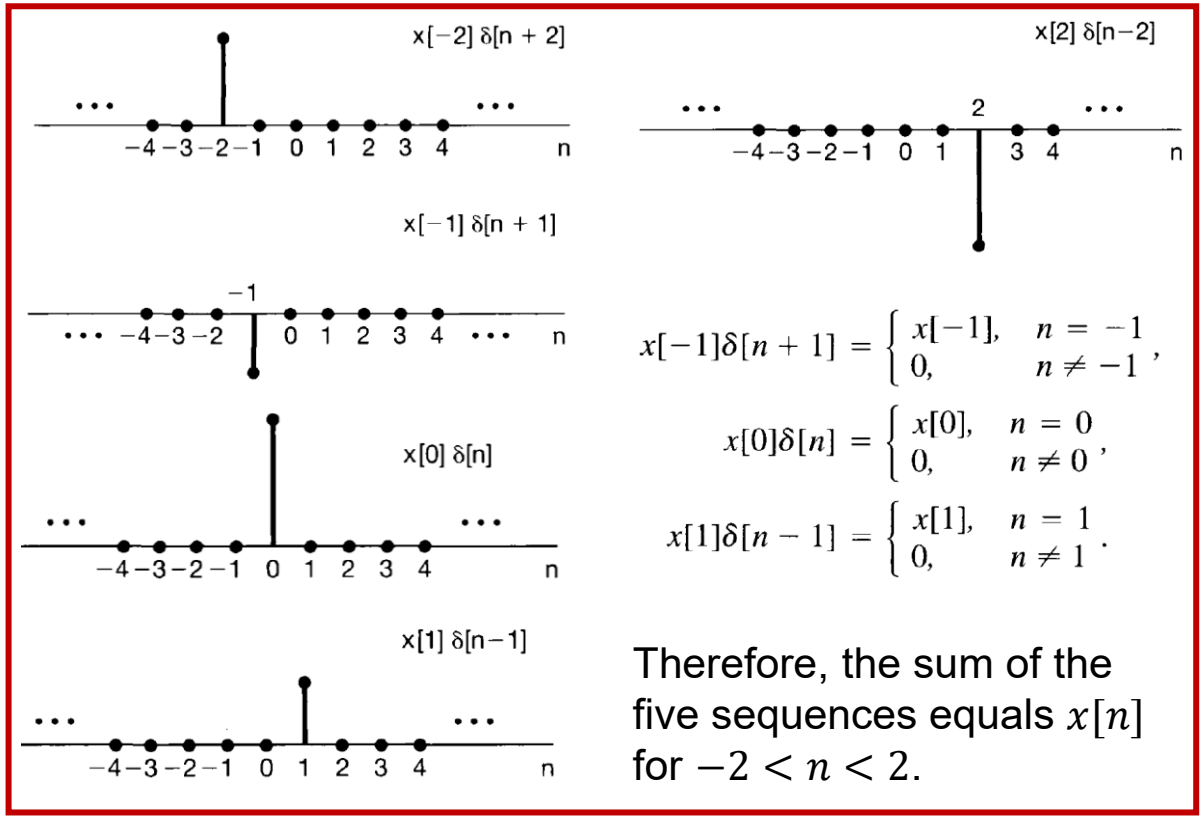
If we know $h[n]$, we can calculate the output for ANY input using **convolution**.

Representation of Discrete-Time Signals in Terms of Impulses

Consider the signal $x[n]$ depicted in the figure below.



Five time-shifted, scaled unit impulse sequences are shown, where each impulse equals the value of $x[n]$ at the particular instant the unit sample occurs.



$$x[-1]\delta[n+1] = \begin{cases} x[-1], & n = -1 \\ 0, & n \neq -1 \end{cases},$$

$$x[0]\delta[n] = \begin{cases} x[0], & n = 0 \\ 0, & n \neq 0 \end{cases},$$

$$x[1]\delta[n-1] = \begin{cases} x[1], & n = 1 \\ 0, & n \neq 1 \end{cases}.$$

Therefore, the sum of the five sequences equals $x[n]$ for $-2 < n < 2$.

Representation of Discrete-Time Signals in Terms of Impulses

More generally, by including additional shifted, scaled impulses, we can write

$$
\begin{aligned}
& x[n] \\
& = \cdots + x[-3]\delta[n+3] + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] \\
& + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3] + \cdots
\end{aligned}
$$

Writing this summation in a more compact form, we have

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

Discrete-Time LTI Systems: The Convolution Sum

The Sifting Property

Any signal $x[n]$ is a sum of scaled, shifted impulses.

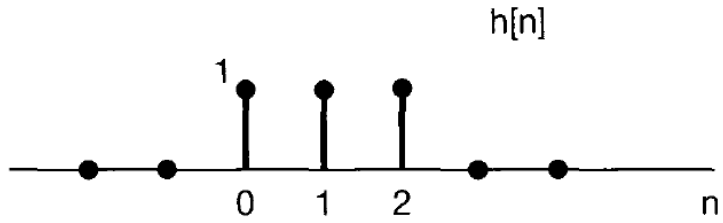
$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

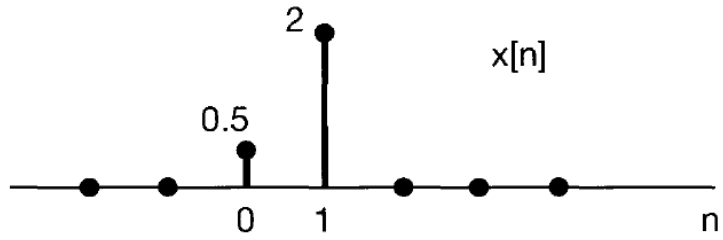
If $h[n]$ is the output of the LTI system when $\delta[n]$ is the input, the response of a linear system to $x[n]$ will be the superposition of the scaled responses of the system to each of these shifted impulses.

$$y[n] = x[n] * h[n] = \underbrace{\sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}_{\text{CONVOLUTION SUM}}$$

Example 2.1

Consider an LTI system with impulse response $h[n]$ and input $x[n]$ as illustrated. Find the output $y[n]$ using the convolution sum.

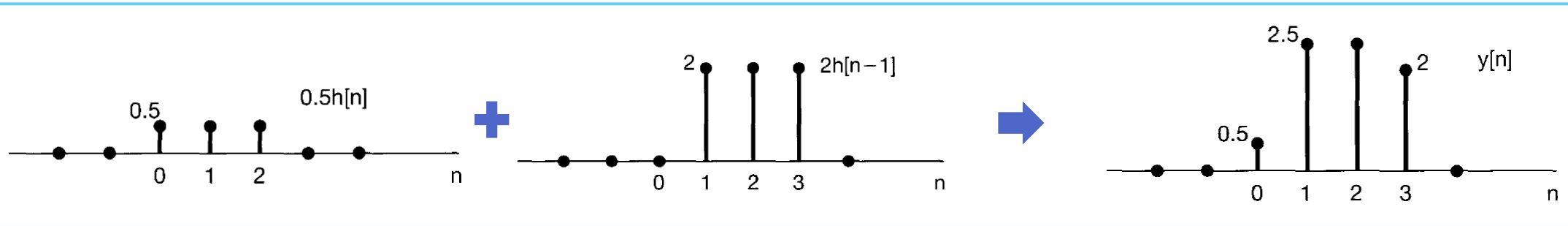




Solution:

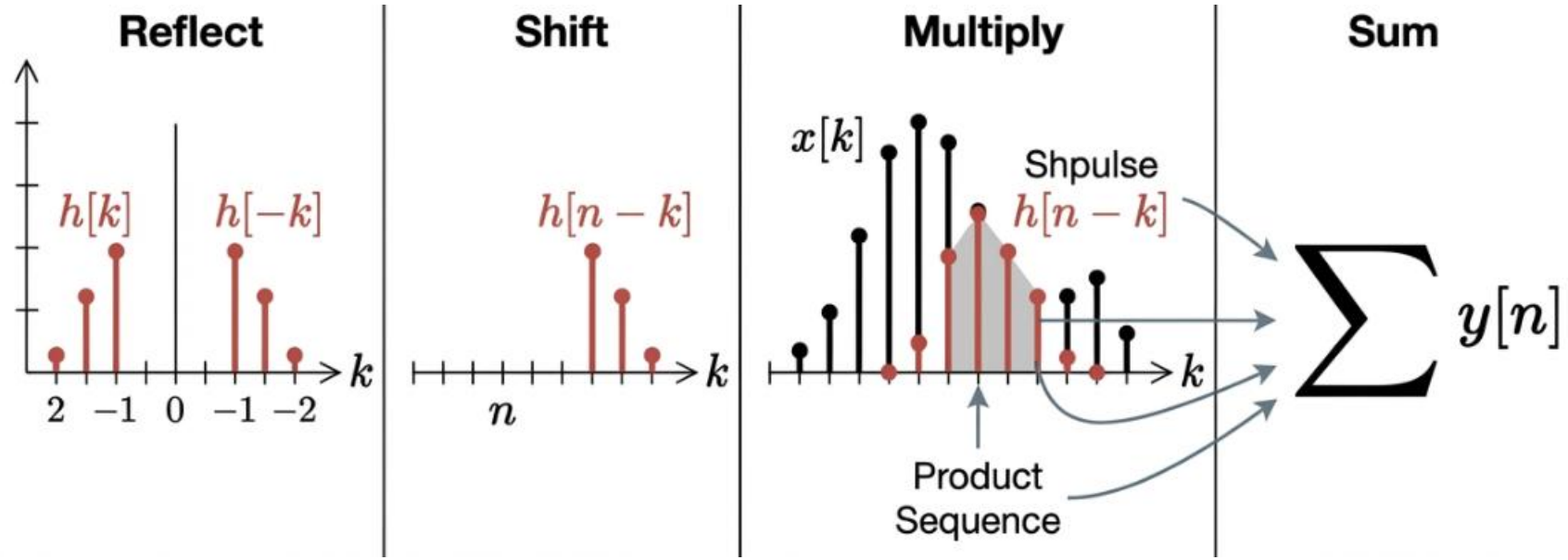
For this case, since only $x[0]$ and $x[1]$ are nonzero, equation $y[n] = x[n] * h[n]$ simplifies to the expression

$$y[n] = x[0]h[n-0] + x[1]h[n-1]$$
$$= 0.5h[n-0] + 2h[n-1]$$



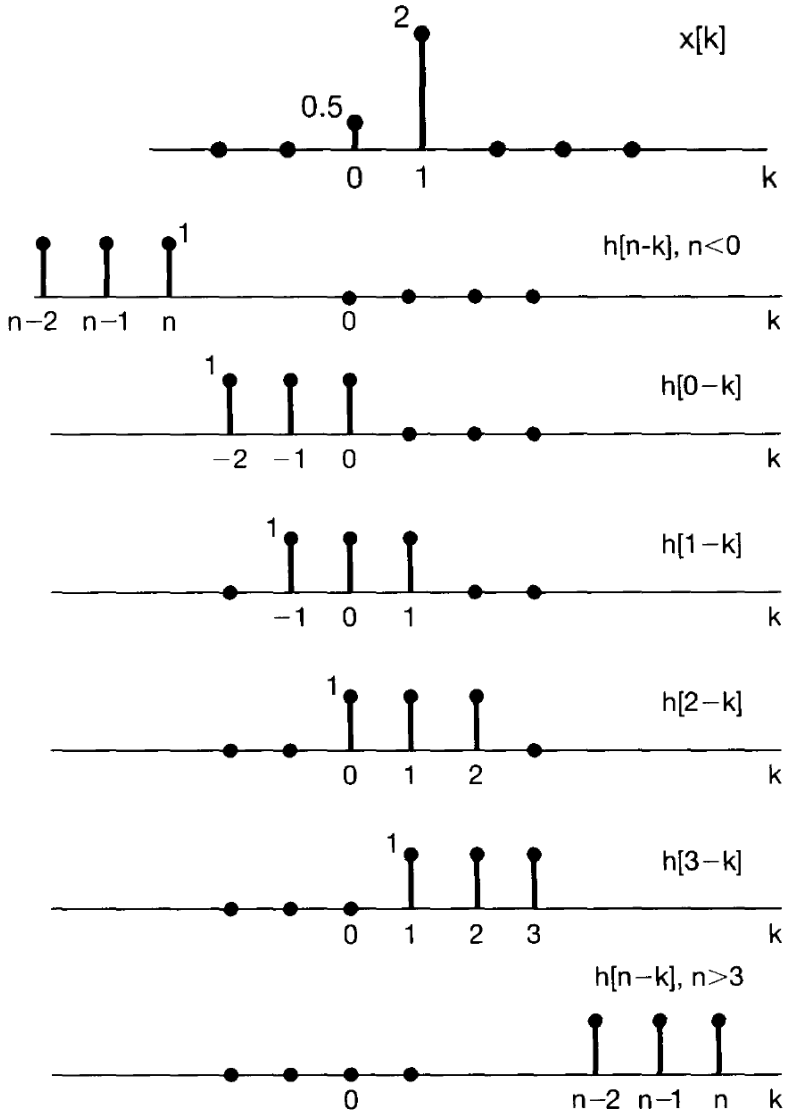
Visualizing Convolution: The Algorithm

Computing $y[n]$ for a specific time n



To compute the output, we reverse the impulse response, slide it across the input, multiply overlapping values, and sum the result.

Example 2.1 – Alternative Method



To compute the output $y[n]$:

1. Reflect impulse response to get $h[-k]$
2. Shift to get $h[n-k]$
3. Multiply overlapping values
4. Sum the result

$y[n] = 0$ for $n < 0$

$$y[0] = \sum_{k=-\infty}^{\infty} x[0]\, h[0-k] = 0.5$$

$$y[1] = \sum_{k=-\infty}^{\infty} x[1]\, h[1-k] = 0.5 + 2 = 2.5$$

$$y[2] = \sum_{k=-\infty}^{\infty} x[2]\, h[2-k] = 0.5 + 2 = 2.5$$

$$y[3] = \sum_{k=-\infty}^{\infty} x[3]\, h[3-k] = 2$$

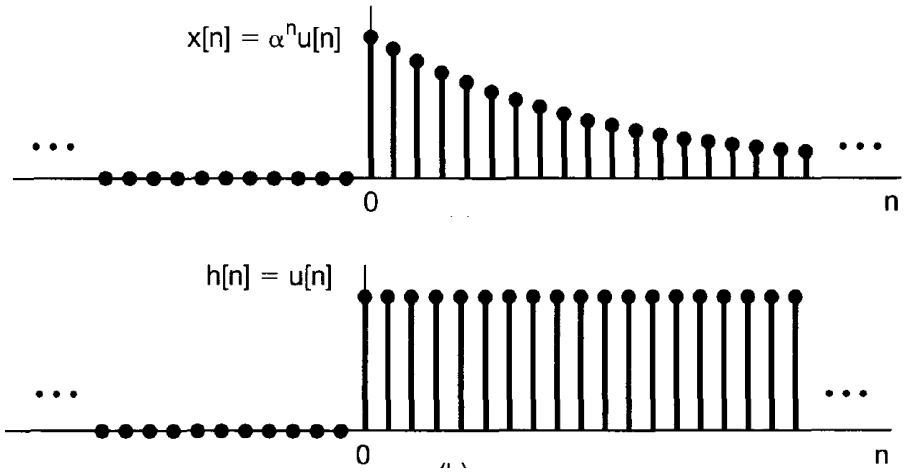
$y[n] = 0$ for $n > 3$

Example 2.2

Consider an input $x[n]$ and a unit impulse response $h[n]$ given by

$$x[n] = \alpha^n u[n]$$
$$h[n] = u[n]$$

with $0 < \alpha < 1$. Find $y[n]$.



Solution:

$$x[k]h[n-k] = \begin{cases} \alpha^k, & 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$$

For $n \ge 0$,

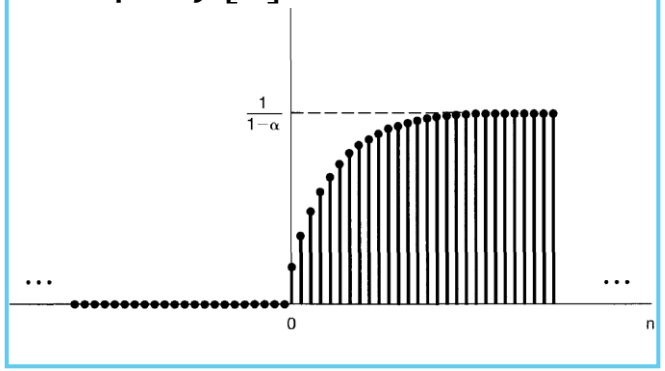
$$y[n] = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}$$

For all n,

$$y[n] = \left(\frac{1-\alpha^{n+1}}{1-\alpha}\right) u[n]$$

← finite sum

Output $y[n]$:



Infinite Sum Formula

$$\sum_{k=0}^{\infty} \alpha^k = \frac{1}{1-\alpha}, \qquad |\alpha| < 1$$

Finite Sum Formula

$$\sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha} \quad \text{for } n \ge 0$$

Example 2.3

Consider an LTI system with input $x[n]$ and unit impulse response $h[n]$ specified as follows:

$$x[n] = 2^n u[-n]$$
$$h[n] = u[n]$$

Find $y[n]$.

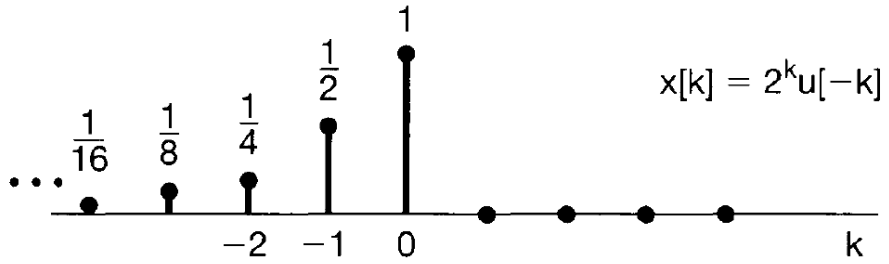
Solution:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Input $x[k] = 2^k u[-k]$:

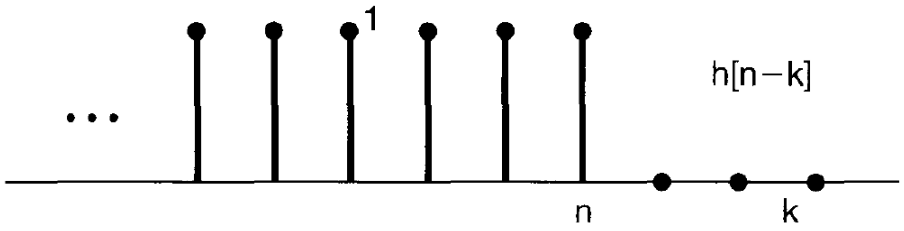
$u[-k] = 1$ for $k \leq 0$

$\Rightarrow x[k] = 2^k$ for $k \leq 0$



Impulse response $h[n-k] = u[n-k]$:

$h[n-k] = u[n-k] = 1$ for $k \leq n$



Example 2.3 (cont'd)

Regardless of the value of n, the sequence $x[k]h[n-k]$ always has nonzero samples along the k-axis.

The product is nonzero when both conditions hold: $k \leq 0$ and $k \leq n$

So the effective upper limit is: $k \leq \min(0, n)$

Case 1: $n \geq 0$ $(\min(0, n) = 0)$

$$y[n] = \sum_{k=-\infty}^{0} x[k]\, h[n-k] = \sum_{k=-\infty}^{0} 2^k$$

Let $k = -r$,

$$y[n] = \sum_{r=0}^{\infty} \left(\frac{1}{2}\right)^r = \frac{1}{1 - {}^1/_2} = 2$$

Infinite Sum Formula

$$\sum_{k=0}^{\infty} \alpha^k = \frac{1}{1-\alpha}, \qquad |\alpha| < 1$$

Finite Sum Formula

$$\sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha} \quad \text{for } n \geq 0$$

Example 2.3 (cont'd)

Case 2: $n < 0$ $(\min(0, n) = n)$

$$y[n] = \sum_{k=-\infty}^{n} x[k]\, h[n-k] = \sum_{k=-\infty}^{n} 2^k$$

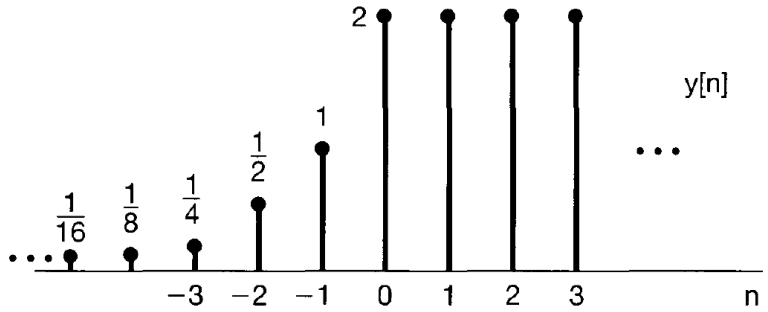
Let $k = n - m$,

$$y[n] = \sum_{m=0}^{\infty} 2^{n-m} = 2^n \sum_{m=0}^{\infty} \left(\frac{1}{2}\right)^m$$

$$y[n] = 2^n \cdot \frac{1}{1 - {}^1/_2} = 2^{n+1}$$

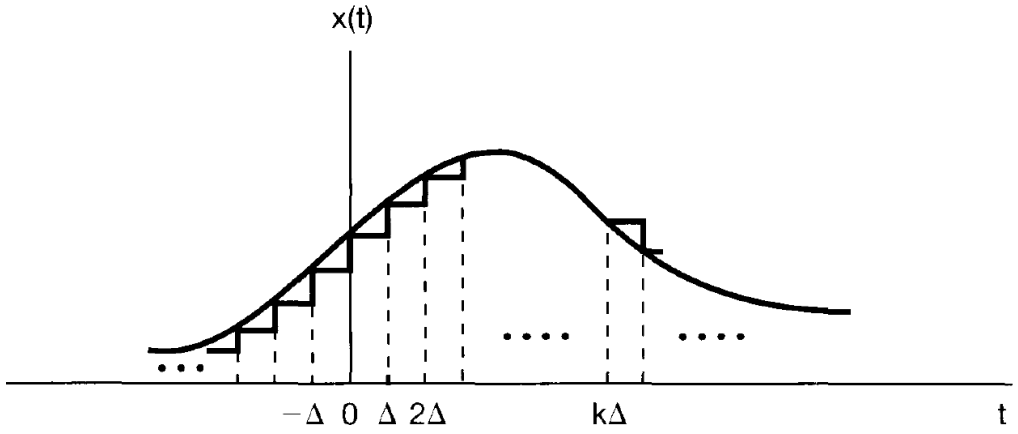
Therefore,

$$y[n] = \begin{cases} 2, & n \geq 0 \\ 2^{n+1}, & n < 0 \end{cases}$$



Continuous-Time LTI Systems: The Convolution Integral

We begin by considering a pulse or "staircase" approximation, $\hat{x}(t)$ to a continuous-time signal, $x(t)$.



$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_{\Delta}(t - k\Delta)\Delta$$

As $\Delta \to 0$, the summation approaches an integral and $\delta_{\Delta}(t)$ is the unit impulse function $\delta(t)$.

The Sifting Property

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)d\tau$$

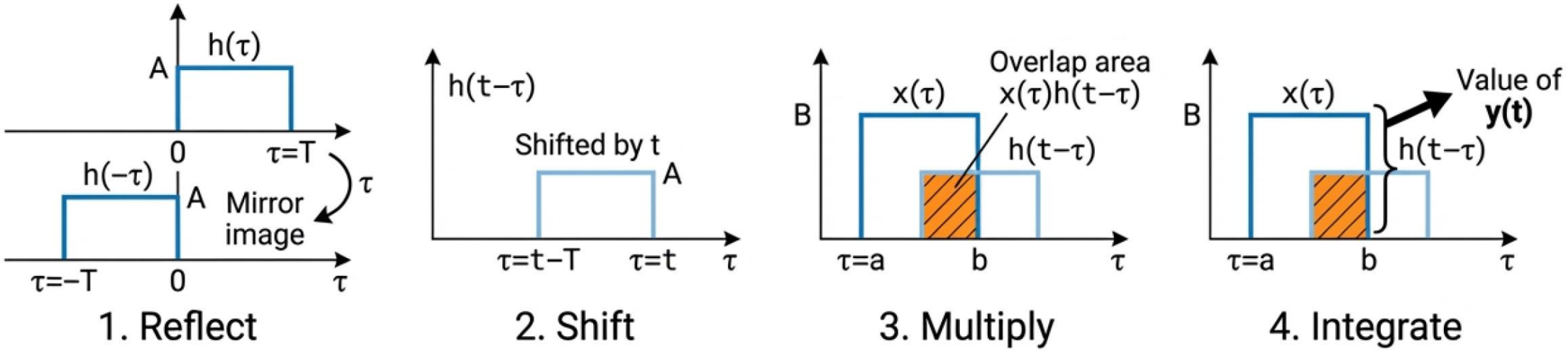
$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

CONVOLUTION INTEGRAL

Visualizing Convolution: The Algorithm

Computing $y(t)$ for a specific time t

For any value of t, the output $y(t)$ is a weighted integral of the input, where the weight on $x(t)$ is $h(t-\tau)$.



To compute the output, we reverse the impulse response, slide it across the input, multiply overlapping values, and integrate the result.

Example 2.4

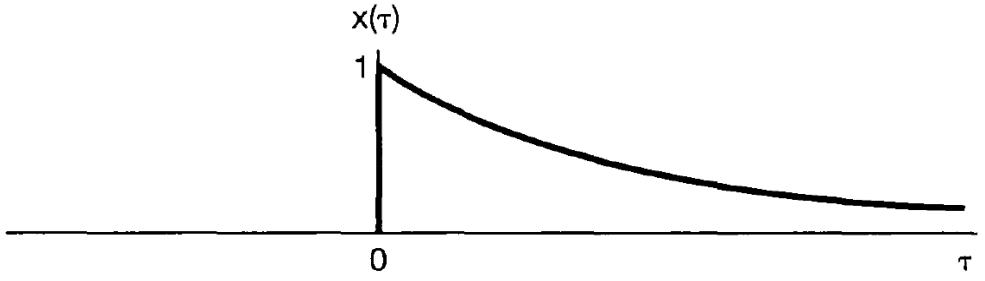
Let $x(t)$ be the input to an LTI system with unit impulse response $h(t)$, where

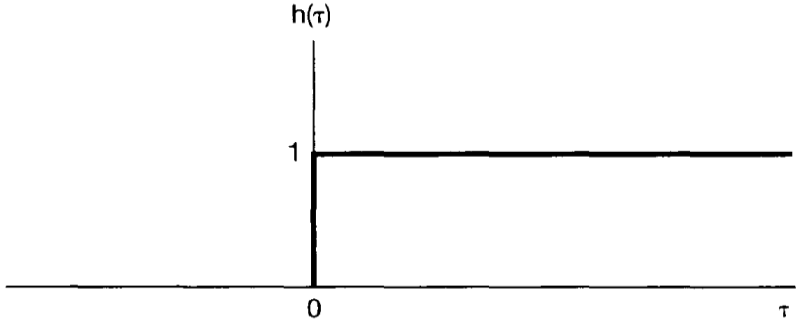
$$x(t) = e^{-at}u(t),\ a > 0 \qquad \text{and} \qquad h(t) = u(t)$$

Find the output $y(t)$.

Solution:

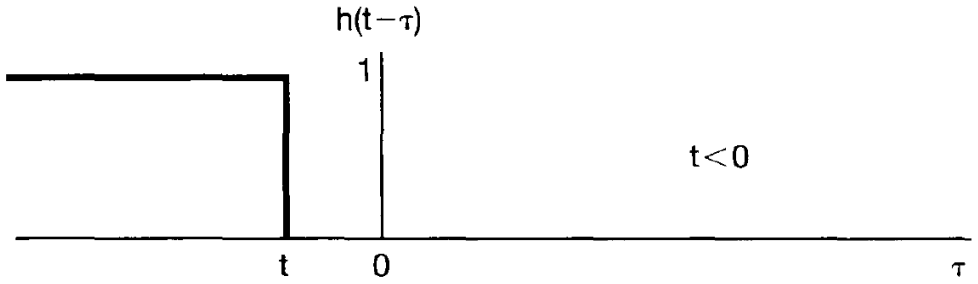
We begin by plotting the functions x and h.

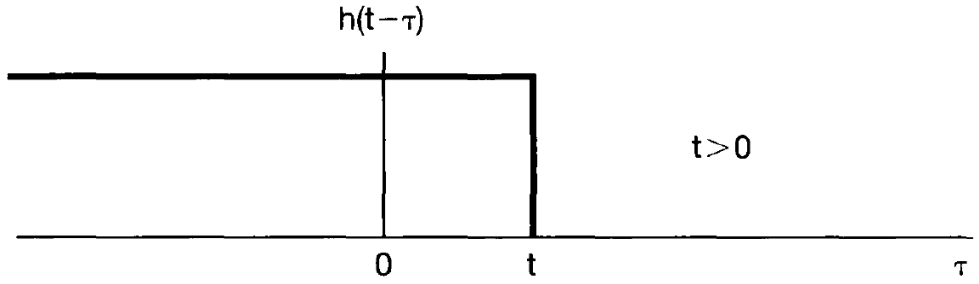




Example 2.4 (cont'd)

Next, we proceed to determine the time-reversed and time-shifted version of $h(\tau)$.





At this point, we are ready to begin considering the computation of the convolution integral. For each possible value of t, we must multiply $x(\tau)$ by $h(t-\tau)$ and integrate the resulting product with respect to t.

Example 2.4 (cont'd)

For $t < 0$, the product of $x(\tau)$ and $h(t-\tau)$ is zero, and consequently, $y(t)$ is zero.

For $t > 0$,

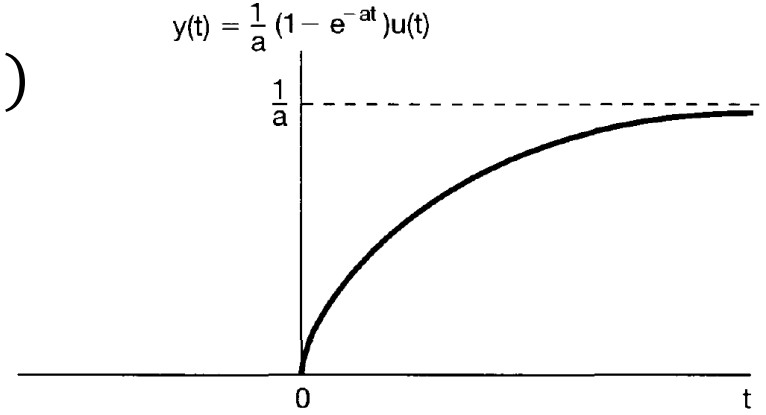
$$x(\tau)h(t-\tau) = \begin{cases} e^{-a\tau}, & 0 < \tau < t \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = \int_0^t e^{-a\tau} d\tau = -\frac{1}{a} e^{-a\tau}\Big|_0^t$$

$$= \frac{1}{a}(1 - e^{-at})$$

Thus, for all t,

$$y(t) = \frac{1}{a}(1 - e^{-at})u(t)$$



Example 2.5

Find the convolution of the following two signals:

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$



$$h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$$

Solution:

Determine the limits of integration

The convolution of two continuous-time signals, $x(t)$ and $h(t)$, is defined by the integral:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Example 2.5 (cont'd)

From the problem statement, we have:

- $x(\tau) \neq 0$ only for $0 < \tau < T$
- $h(t-\tau) \neq 0$ only when $0 < t - \tau < 2T$

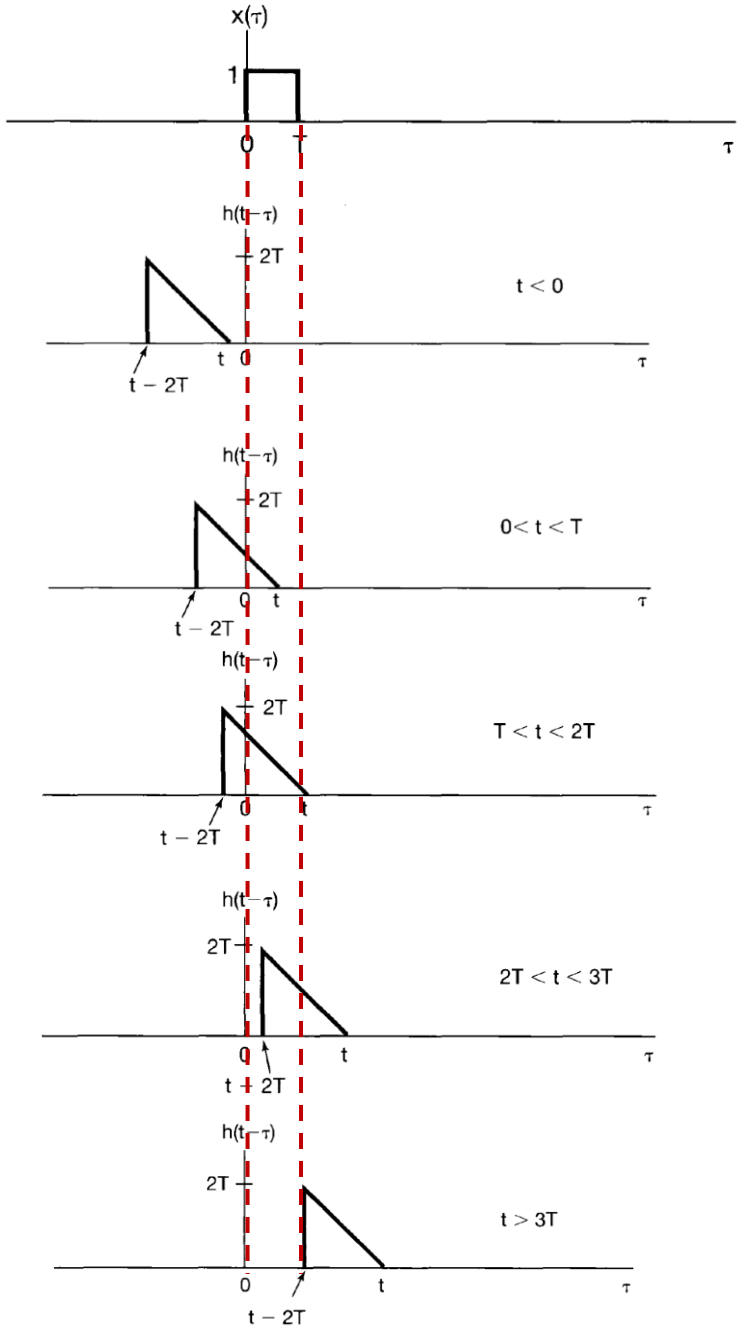
  $\Rightarrow t - 2T < \tau < t$

$\tau \in [\max(0, t-2T), \min(T, t)]$

Case 1: $0 < t < T$ (Overlap: $0 < \tau < t$)

$$y(t) = \int_0^t 1 \cdot (t-\tau) d\tau$$

$$y(t) = \frac{1}{2} t^2$$



Example 2.5 (cont'd)

Case 2: $\boldsymbol{T < t < 2T}$ (Overlap: $0 < \tau < T$)

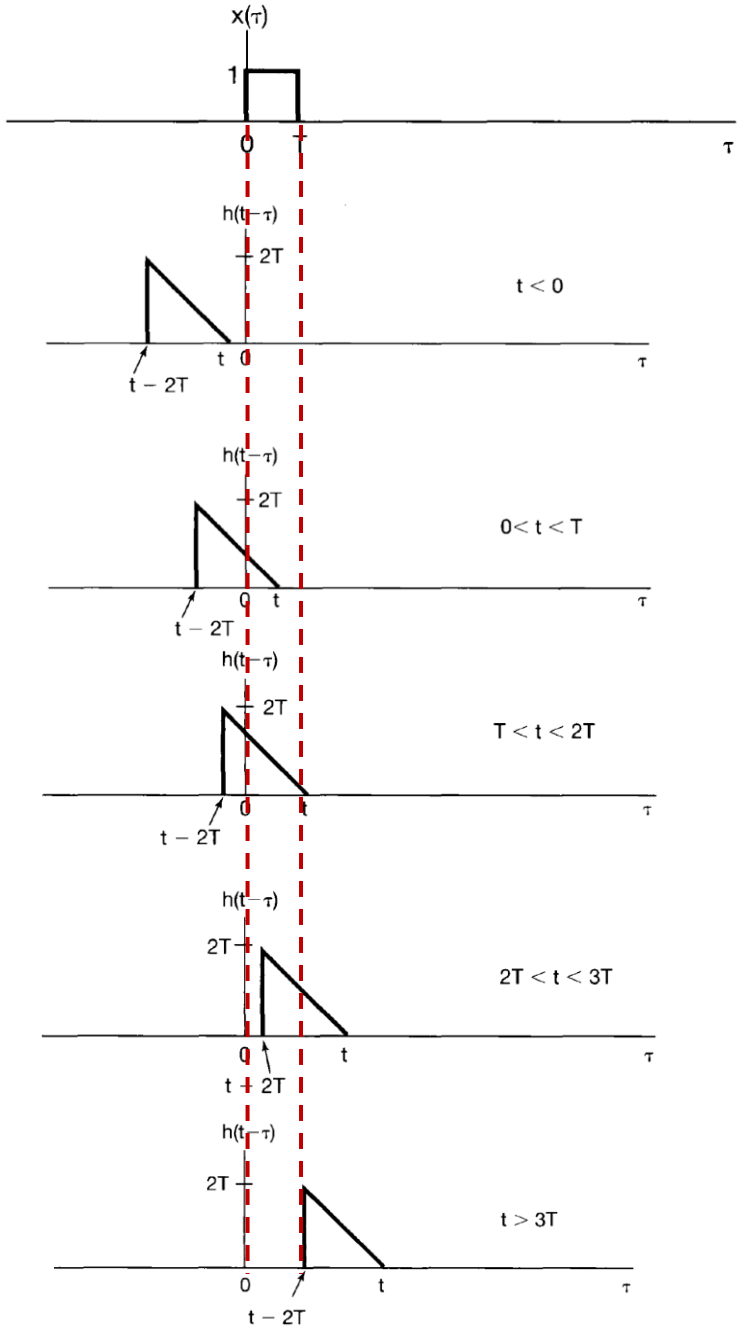
$$y(t) = \int_0^T 1 \cdot (t - \tau) d\tau$$

$$y(t) = Tt - \frac{1}{2}T^2$$

Case 3: $\boldsymbol{2T < t < 3T}$ (Overlap: $t - 2T < \tau < T$)

$$y(t) = \int_{t-2T}^T 1 \cdot (t - \tau) d\tau$$

$$y(t) = -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2$$



Example 2.5 (cont'd)

Therefore,

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}t^2, & 0 < t < T \\ Tt - \frac{1}{2}T^2, & T < t < 2T \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2, & 2T < t < 3T \\ 0, & t > 3T \end{cases}$$

